

Kareem Jaber

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EDUCATION

Princeton University

A.B. in Mathematics, GPA 3.975

Expected minor in Computer Science

Princeton, NJ

Aug. 2022 – Present

PUBLICATIONS, PREPRINTS, AND INDEPENDENT WORK

Expository Papers

- [On Dimension-Free \$L^p\$ bounds for Maximal Functions](#): My junior independent work at Princeton, advised by Assaf Naor. I wrote an exposition of work of Stein, Bourgain, Carbery, and Muller on dimension independent estimates for the Hardy-Littlewood maximal functions of families of convex bodies.
- [A Proof of the Gauss-Bonnet Theorem](#): My final paper for a Junior Seminar on Geometric Analysis, advised by Ravi Shankar.

Research Papers

- [Congruence Classes of Simplex Structures in Finite Field Vector Spaces](#): at SMALL REU 2024, with Tim Cheek, Joe Cooper, Pico Gilman, Vismay Sharan, Jenna Shuffelton, and Marie-Hélène Tomé, advised by Eyvindur Palsson and Alex Iosevich. Accepted in the *Bulletin of the Hellenic Math Society*.
- [Numerical Investigation of Lower Order Biases in Moment Expansions of One Parameter Families of Elliptic Curves](#): at SMALL REU 2024, with Tim Cheek, Pico Gilman, Marie-Hélène Tomé, and Vismay Sharan, advised by Steven J. Miller. Accepted in the *Journal of Number Theory*, Volume 279.
- [On the Density of Low Lying Zeros of a Large Family of Automorphic \$L\$ -functions](#): at SMALL REU 2024, with Tim Cheek, Pico Gilman, and Marie-Hélène Tomé, advised by Steven J. Miller.
- [Colorings of Symmetric Unions and Partial Knots](#): At the Polymath Junior REU 2022 (virtual). With Ben Clingenpeel, Zongzheng Dai, Gabriel Diraviam, Krishnendu Kar, Ziyun Liu, Teo Micklethun, Haritha Nagampoozhy, Michael Perry, Moses Samuelson-Lynn, Eli Seamans, Ana Wright, Nicole Xie, and Ruiqi Zou, advised by Alexander Zupan. Accepted in the *Journal of Knot Theory and its Ramifications*.

Recreational Mathematics Papers

- [Variants of Conway Checkers and k-nacci Jumping](#): At SMALL REU 2024, with Glenn Baruda, Joe Cooper, and Raul Marquez, advised by Steven J. Miller. Accepted in *The Fibonacci Quarterly*.
- [The Unsolvable Configuration of the 15 Puzzle and an Interesting Approach to Abstract Algebra](#): An expository paper I wrote right before entering university, which won runner-up for the Steven H. Strogatz Prize for Math Communication 2022.

INVITED TALKS

- Upper Bounds for the Lowest First Zero in Families of Cuspidal Newforms (presented with Glenn Bruda, Timothy Cheek, Steven J. Miller, Vismay Sharan and Saad Waheed), AMS Special Session on L -functions, Automorphic Forms, and Their Applications, Joint Math Meetings, Seattle, January 11, 2025.
- Congruence Classes of Simplex Structures in Finite Field Vector Spaces (presented with Vismay Sharan), PME Contributed Session on Research by Undergraduates VII, Joint Math Meetings, Seattle, January 10, 2025.
- Congruence Classes of Simplex Structures in Finite Field Vector Spaces (presented with Vismay Sharan), Young Mathematicians Conference, The Ohio State University, August 15, 2024.

RELEVANT EXPERIENCE

Senior Thesis | Princeton, NJ

Fall 2025 - Present

- Studying complex algebraic geometry under Prof. János Kollár, and expositing a proof of the Enriques-Kodaira classification of compact complex surfaces

Geometric Measure Theory Research | Princeton, NJ

Summer 2025

- Worked under Professors Camillo De Lellis and Federico Glaudo on improving best known results on the Besicovitch $1/2$ conjecture, in both computational and theoretical directions

Algebraic Geometry Independent Work | Princeton, NJ

Summer 2025

- Worked under Jakub Witaszek in learning the foundations of scheme theory and sheaf cohomology through Vakil's *The Rising Sea*, in preparation for my senior thesis

SMALL Williams REU | Williamstown, MA

Summer 2024

- Researched point configurations over finite fields with Professors Eyvindur Palsson and Alex Iosevich, leading to a paper and talks at the 2024 Young Mathematicians Conference as well as the 2025 Joint Mathematics Meetings
- Researched the distribution of low-lying zeros of L -functions with Prof. Steven J. Miller, resulting in a paper and a talk at the 2025 Joint Mathematics Meetings
- Computationally investigated lower order biases in moment expansions of one parameter families of elliptic curves with Prof. Steven J. Miller, resulting in a publication in the *Journal of Number Theory*

Undergraduate Course Assistant | Princeton University

Sept. 2023 - Present

- Work as a course assistant for Princeton's advanced introductory analysis sequence (MAT216/218) and introductory algebra course (MAT345)
- Write companion notes for the course material to aid students' understanding and intuition

Engineering Teacher | Kuching, Malaysia; International Internship Program

Summer 2023

- Worked at Chumbaka, an engineering education company in Malaysia; co-taught their coding and electronics curriculum in after-school programs; trained teachers in using material in their classrooms; contributed to company-wide curriculum development and R&D meetings
- Mentored a group of secondary students designing a device to track hemorrhagic shock treatment information in a hospital environment; students won first place in regional competition and are now working closely with Sarawak emergency room doctors

Polymath Jr REU | Virtual

Summer 2022

- Collaborated virtually for 8 weeks under the guidance of Prof. Alexander Zupan, University of Nebraska-Lincoln, focusing on n -colorings of symmetric union presentations, resulting in a publication in the *Journal of Knot Theory and It's Ramifications*

Mathematics and Education Outreach

Summer 2020 - present

- Established my own math circle after-school program, the Princeton Community Math Circle, at two local elementary schools in the Princeton area
- Attended the Math Circle Institute 2025 at Notre Dame to share my outreach experiences and learn from professional groups involved with mathematics outreach globally
- Organize and administer Princeton Splash, an annual education outreach event for high school students, and represented Princeton at SplashCon 2024, a conference at MIT to learn more about effective education outreach
- Write expository mathematics papers on problem solving and undergraduate mathematics for high schoolers

SELECTED AWARDS AND HONORS

Princeton University Andrew H. Brown Prize 2025, awarded to four outstanding juniors in mathematics

HackPrinceton 2025 Education Category Winner for a 4D VR graphing calculator

William Lowell Putnam Mathematical Competition 2023, Top 500 nationally

Princeton University Shapiro Prize For Academic Excellence 2023 (awarded to top 3% of students in class)

Princeton International Internship Program 2023, Best Reflection

Runner up in Steven H. Strogatz Prize for Math Communication 2022